

Ahmed Usman

AI/ML Engineer

📞 +92 335 0707006 | ✉️ ahmadusman050@gmail.com |
🐙 github.com/ahmedosmo | 🔗 linkedin.com/in/ahmedusman050

Objective

AI/ML Engineer with expertise in deep learning, LLM applications and MLOps. Experienced in developing NLP and speech processing systems, with proficiency in RAG, Agentic AI, Fine-tuning and cloud deployment. Proven track record of delivering optimized, production-ready AI solutions.

Skills

Programming Languages: Python, JavaScript, C++

Web Frameworks: Flask, FastAPI, Node.js, Express.js

Databases: MongoDB, MySQL, Postgres

Data Engineering & Processing: NumPy, Pandas, Matplotlib, Seaborn

AI & Automation: TensorFlow, Scikit-learn, N8N

Natural Language Processing: LangChain, LangGraph, CrewAi, Vector Databases

Speech Processing: SciPy, Librosa, Coqui, Whisper, Retell AI, ElevenLabs

MLOps: Docker, Jenkins, Git, GitHub, CI/CD, LangSmith

Soft Skills: Communication, Team Collaboration, Problem-Solving, Fast Learning

Languages: English(C1), German(A2)

Experience

Associate AI Engineer at Tech Emulsion - (full time) (Jul, 2025 - Present)

- Design and deploy LLM-driven conversational agents, RAG pipelines, and automation workflows by integrating APIs, structured backends, and third-party tools to deliver scalable AI solutions.
- Develop and maintain workflows using Python, Django, FastAPI, n8n, and AWS, while collaborating with cross-functional teams to deliver end-to-end AI products.

AI/ML Engineer at DevK System - (part time) (Aug, 2024 - Jun, 2025)

- Designed and Developed LLM Applications including RAG Systems, Agentic AI and Autonomous Agents.
- Designed and Deployed ML models, ensuring scalability and low latency.
- Applied MLOps practices with Docker, CI/CD, and model monitoring.

Back-End Developer at Brandora - (part time) (Sep, 2022 - Jul, 2024)

- Developed REST APIs with Node.js, Express.js, and MongoDB.
- Integrated APIs with React front-end for seamless data flow.
- Implemented authentication and optimized database queries.

Projects

Speech Processing & Voice Agents

- [Speech Enhancement \(FYP\)](#) – Built a U-Net + BiLSTM model on Voice Bank & DEMAND dataset, achieving gains (PESQ +8.45%, STOI +1.56%, ESTOI +12.05%), with a Gradio interface and published research.
- [Multi-Agent Voice Automation System \(E-commerce\)](#) – Built 4 AI voice agents (Prospecting, Abandoned Cart, Winback, Inbound Support) using Retell AI, n8n, and nodejs integrated Shopify APIs for real-time customer data and automated sales recovery.
- [AI Health Receptionist Voice Agent](#) – Developed an ElevenLabs-based healthcare voice agent enabling appointment booking, cancellation, and rescheduling with payment integration across PMS platforms (Dentally, Pabau).
- [Coqui TTS – Text-to-Speech](#) – Developed a Coqui TTS + Gradio app for text-to-speech with GPU/CPU support, real-time playback, and audio download.
- [Silence Removal from Audio](#) – Created a Streamlit app to remove silent audio segments with adjustable thresholds, waveform visualization, and playback/download options.

[LLM Based Applications \(LLM, RAG, and Agent Systems\)](#)

- [Conversational RAG PDF Chatbot](#) – Multi-turn Q&A chatbot with LangChain, Groq API, HuggingFace embeddings, ChromaDB for semantic search; accurate, context-aware answers from PDF documents.
- [Multi-Source Search Agent](#) – Integrated Wikipedia, Arxiv, and DuckDuckGo into a LangChain Agent with Groq API + LLaMA3-8B for multi-source conversational retrieval.
- [SQL Chatbot](#) – Natural language queries to SQLite & MySQL with LangChain Agents and Groq API, dynamic database selection via Streamlit UI.
- [Content Extraction & Summarizer](#) – Extracted & summarized YouTube videos and web pages with YoutubeLoader, UnstructuredURLLoader, and custom summarization prompts.
- [Text-to-Math Solver & Data Assistant](#) – Combined LLMMathChain and Wikipedia API with custom reasoning prompts to handle math, factual queries, and logic-based tasks.
- Technologies: Python, LangChain, LangGraph, LangSmith, Groq, HuggingFace, VectorDB, SQLite, MySQL, Ollama, OpenAI, Docker

Education

University of Engineering and Technology, Peshawar

B.S. Electrical Computing & Communication Engineering

Certifications

- [Machine Learning Specialization](#)
- [Deep Learning Specialization](#)
- [Machine Learning in Production](#)
- [TensorFlow for AI, ML and DL](#)
- [Complete Generative AI Course With Langchain and Huggingface](#)